

Please check the examination details below before entering your candidate information		
Candidate surname	Other names	
Centre Number	Candidate Number	
Pearson Edexcel Level 3 GCE		Paper reference 9EB0/03
Predicted June 2026 Morning (Time: 2 hours)		
Economics B Advanced PAPER 3: The economic environment and business		
You must have: Insert (enclosed)		Total Marks

PREDICTED PRACTICE PAPER — ADVANCED

Pre-release context for this paper: innovation, technology and the distribution of economic power

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided — *there may be more space than you need.*

Information

- The total mark for this paper is **100**.
- The marks for **each** question are shown in brackets — *use this as a guide as to how much time to spend on each question.*
- Calculators may be used.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Answer ALL questions.

SECTION A

Read the following extracts (A to D) before answering the question.

Write your answers in the spaces provided.

Extract A

The global semiconductor industry

Semiconductors are essential inputs to virtually every modern manufactured good — from smartphones and cars to medical devices, defence equipment and the data centres that power artificial intelligence (AI). The global semiconductor market was estimated at around US\$630 billion in 2024 and is forecast to exceed US\$1 trillion by 2030.

Semiconductor manufacturing is extraordinarily capital-intensive: a single leading-edge fabrication plant ('fab') costs US\$15–30 billion to build and takes three to five years to bring on-line. TSMC (Taiwan) produces approximately 60% of the world's foundry output and over 90% of chips at the 5nm node or below.

ASML (Netherlands) is the sole global producer of extreme-ultraviolet (EUV) lithography machines. Electronic design automation (EDA) software is dominated by three US firms; design IP is dominated by US and UK firms (including ARM).

(Source: adapted from SIA and industry reports, 2024)

Extract B

US export controls and the CHIPS Act

In October 2022, the US Department of Commerce announced sweeping export controls restricting the sale of advanced chips, chip-making equipment and US citizens' services to China. The controls were expanded in October 2023 and again in December 2024.

The US CHIPS and Science Act (2022) committed approximately US\$52 billion of federal funding and US\$24 billion of investment tax credits to rebuild domestic semiconductor manufacturing, precipitating announcements of over US\$450 billion in private-sector investment commitments. The European Union passed a similar €43 billion EU Chips Act in 2023; China's Big Fund phase III mobilised a further US\$47.5 billion in 2024.

(Source: adapted from US Commerce Department and EU Commission, 2024)

Extract C

Geopolitics and supply-chain concentration

The concentration of leading-edge chip manufacturing in Taiwan — an island of 23 million people located less than 200 km from mainland China — is widely viewed as the most acute single point of failure in the global economy. A prolonged interruption to Taiwanese chip production would, according to a 2023 Rhodium Group estimate, reduce global GDP by several trillion US dollars within a year.

Even absent geopolitical disruption, the industry has experienced marked cyclicity. A cycle of over-ordering followed the 2020–21 Covid-linked shortages, producing a memory-chip glut in 2023 during which prices of DRAM fell by over 60%.

(Source: adapted from Rhodium Group and industry data, 2024)

Extract D

AI demand and Nvidia

In 2023–24, generative AI drove unprecedented demand for high-end graphics processing units (GPUs). Nvidia, the US designer of the dominant H100 and H200 data-centre GPUs, saw its annual revenue rise from US\$27bn (FY23) to US\$61bn (FY24) to an expected US\$125bn+ (FY25). Its market capitalisation briefly exceeded US\$3 trillion in 2024.

Nvidia's chips are designed in the US, manufactured by TSMC in Taiwan, packaged in Taiwan and Korea, and deployed in data centres globally. US export controls restrict the sale of Nvidia's most advanced products to China; Nvidia designed 'compliant' variants (A800, H20) but these were subsequently restricted in later rounds of controls.

(Source: adapted from Nvidia 10-K filings, 2024)

Region	1990	2010	2020	2024
Taiwan	≈0	17	22	23
South Korea	≈0	17	18	17
Mainland China	≈0	9	15	19
Japan	49	20	14	12
USA	37	13	12	11
Europe (EU + UK)	14	10	9	9
Rest of World	≈0	14	10	9

Figure 1: Global semiconductor market — share of global production by region (%)

Region	USA	Taiwan	S.Korea	Japan	EU	China
Announced capex	223	148	138	63	51	142

Figure 2: Announced capital expenditure (US\$ bn) — new semiconductor fabs, 2022–2027

Metric	FY22	FY23	FY24	FY25 (est.)
Revenue	26.9	27.0	60.9	125.0
Gross margin (%)	64.9	56.9	72.7	75.0
Operating profit	10.0	4.2	32.0	82.0
R&D expenditure	5.3	7.3	8.7	12.0
Year-end market cap (\$tr)	0.60	0.37	1.22	3.30

Figure 3: Nvidia selected financials (US\$ bn, fiscal years ending Jan)

(8)

(b) With reference to Extracts B and C and Figures 1 and 2, discuss the macroeconomic consequences for the United States of the concentration of semiconductor production in East Asia.

(10)

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(d) Using Extracts B, C and D and your own knowledge, evaluate whether the ongoing 'technology decoupling' between the United States and China is likely to accelerate or hinder global economic growth over the next decade.

(20)

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(Total for Question 1 = 50 marks)
TOTAL FOR SECTION A = 50 MARKS

Answer ALL questions.

SECTION B

Read the following extracts (E to H) before answering the question.

Write your answers in the spaces provided.

Extract E

The rise of generative AI and a dominant stack

Since the public release of ChatGPT in November 2022, generative AI has become the most rapidly adopted consumer technology in history. ChatGPT reached 100 million weekly active users within two months.

Three broad layers have emerged in the generative-AI stack: at the bottom, specialised hardware — primarily Nvidia GPUs; in the middle, 'foundation models' trained at vast cost by a small group of developers including OpenAI (partnered with Microsoft), Anthropic (partnered with Amazon and Google), Google DeepMind, Meta and Mistral; at the top, a rapidly expanding applications layer.

(Source: adapted from Stanford AI Index 2024)

Extract F

Scaling costs and energy demand

Foundation-model training costs have risen exponentially. GPT-4 is estimated to have cost approximately US\$100 million to train; analysts estimate that frontier models released in 2027 may cost over US\$10 billion. Inference (running the model) has become the dominant cost for widely-deployed models.

The International Energy Agency estimates that global data-centre electricity consumption could roughly double by 2026, with AI inference accounting for a rapidly growing share. Hyperscalers — Microsoft, Amazon, Google and Meta — have collectively signed long-term contracts for over 30 GW of low-carbon power, including new nuclear, solar and geothermal projects.

(Source: adapted from Epoch AI and IEA reports, 2024)

Extract G

Labour market and productivity effects

Goldman Sachs estimated in 2023 that generative AI could raise global labour productivity by 1.5 percentage points per year for a decade; MIT and Princeton studies of specific tasks have shown productivity gains of 14–56%. The IMF has warned that around 40% of global jobs are 'exposed' to AI, with higher exposure in advanced economies (60%) and in jobs held by women and higher earners.

In early 2024, the EU AI Act was adopted — the world's first comprehensive AI regulation, imposing transparency and risk-management obligations on 'general purpose AI systems', with fines of up to 7% of global turnover for breaches.

(Source: adapted from Goldman Sachs, IMF and European Commission, 2024)

Extract H

Competition and concentration in foundation models

Microsoft has invested over US\$13 billion in OpenAI; Amazon and Google have committed up to US\$8 billion in Anthropic. Critics argue these partnerships represent a new form of concentration: the hyperscaler supplies the compute, the AI firm supplies the model, and the hyperscaler's cloud becomes the primary distribution channel. The EU antitrust regulator (DG COMP), UK's CMA and US FTC have all launched investigations.

In early 2025, DeepSeek — a Chinese AI developer — released an open-weight model that achieved frontier-level performance at a reportedly far lower training cost, briefly wiping US\$600bn from the market capitalisation of US AI-related firms and reopening the question of whether frontier AI is a winner-take-all market.

(Source: adapted from company disclosures and press reports, 2024–25)

Indicator	2022	2023	2024
Global AI private investment (US\$ bn)	91.9	95.8	151.0
Est. training compute, frontier model ($\times 10^{23}$ FLOPs)	2.1	25.0	320
Est. training cost, frontier model (US\$ m)	20	100	500
Hyperscaler capex, combined (US\$ bn)	128	147	235
AI chip sales revenue, all vendors (US\$ bn)	25	52	145

Figure 4: Selected generative-AI economic indicators

(8)

(b) With reference to Extract F, assess the likely macroeconomic consequences of the rise in AI-related data-centre electricity demand for the economies that host major data centres.

(10)

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x} \quad (12)$$

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(d) Evaluate the appropriate role for governments in regulating the market for frontier AI foundation models, using evidence from Extracts E to H. Your answer should consider both microeconomic efficiency and wider macroeconomic consequences.

(20)

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(Total for Question 2 = 50 marks)

TOTAL FOR SECTION B = 50 MARKS

TOTAL FOR PAPER = 100 MARKS

END OF PAPER

INSERT — Pre-release Context

Do not return this Booklet with the question paper.

Investigating the Economic Environment and Business

Context

This year the context is **innovation, technology and the distribution of economic power**. You should think about how new technologies — particularly in semiconductors and generative AI — reshape market structures, national economies, labour markets and the distribution of economic power between firms and between states.

Research

To prepare for this year's context, you should research:

- why certain technology markets exhibit extreme concentration (scale economies, network effects, sunk costs)
- the economic rationale for industrial policy (CHIPS Act, EU Chips Act, China Big Fund)
- how trade and export controls affect global technology supply chains
- the microeconomic and macroeconomic effects of transformative new technologies
- the case for and against government regulation of powerful new technologies.

You **cannot** take any of your research or investigation data carried out as part of the pre-release into the examination.